Week 08: The Frame Problem and Common Sense

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

An intelligent agent must know what changes when it acts and what stays the same, and it must recognise which of its countless beliefs matter. This week traces the frame problem from its logical origin to its philosophical generalisation as a problem of relevance, and connects it with the challenge of common sense [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to distinguish the representational frame problem from the general problem of relevance. They should be able to explain STRIPS and successor state axioms as solutions to the former and to reconstruct Dennett's robot story. Finally, they should be able to explain why Winograd schemas test common sense.

Lecture

A logical problem

McCarthy and Hayes introduced the situation calculus to represent actions and their effects in logic, and noticed a difficulty [1]. To reason that painting a block does not move it, the system needs an explicit axiom saying so. The same holds for every pair of an action and a property it does not affect. The number of these frame axioms grows with the product of actions and properties, called fluents. Figure 8.1 shows the growth. They named this the frame problem.

Two classic answers address the representational side. The STRIPS planner of Fikes and Nilsson represented each action by lists of facts it adds and deletes and assumed that everything else persists [4]. Reiter later showed how to compile all frame axioms into one successor state axiom per fluent, which states exactly when the fluent is true after an action [5]. Shanahan presented a detailed treatment of such solutions based on a common sense law of inertia [6]. In the logical sense, the frame problem is largely solved.

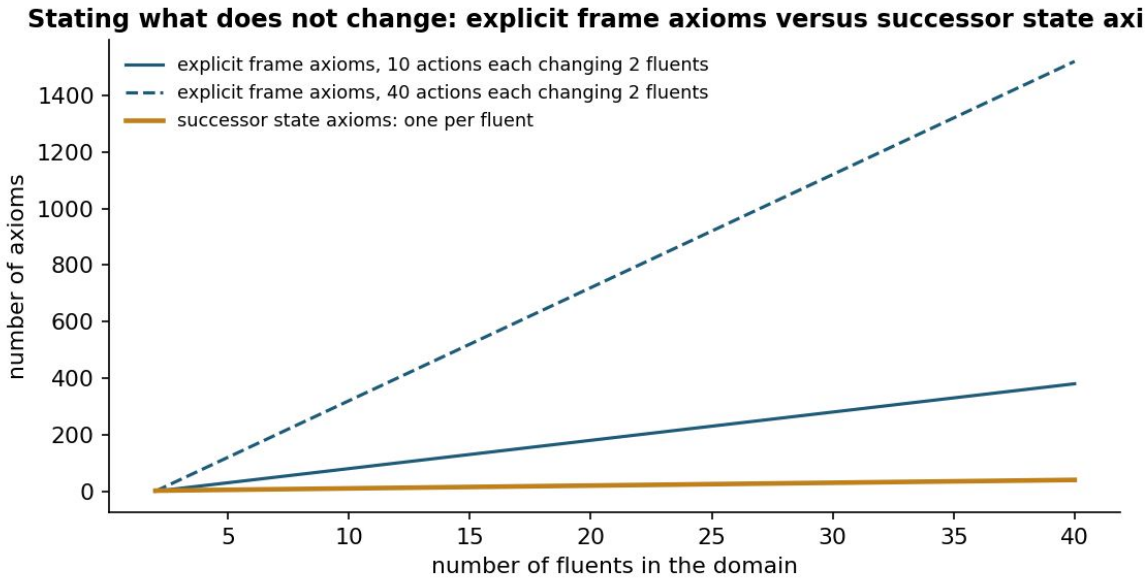


Figure 8.1. Explicit frame axioms grow with the product of actions and fluents, while successor state axioms grow with the number of fluents [1, 5].

A problem of relevance

Dennett told a story that turned the frame problem into a general problem of cognition [2]. A robot, R1, must fetch its spare battery from a room in which a bomb is timed to go off. The battery is on a wagon, and so is the bomb. R1 pulls the wagon out, knowing that the battery is on it, but does not notice that the bomb comes along too. Its designers build R1D1, which deduces the implications of its actions. When the bomb explodes, R1D1 is still deducing irrelevant consequences, such as that pulling the wagon would not change the colour of the walls. The next model, R2D1, is taught to ignore irrelevant implications, but it is busy classifying thousands of them as irrelevant when time runs out.

The moral is that the difficulty lies not in representing change but in determining relevance quickly, without considering everything. People do this effortlessly, and no one knows how. Dreyfus, whose critique is examined in Week 9, saw in the problem evidence that intelligence rests on background coping rather than on rules.

Common sense

Davis and Marcus surveyed the challenge of commonsense reasoning and argued that it remains a central obstacle for AI [3]. Commonsense knowledge is vast, mostly unstated and used without effort: that objects fall when dropped, that containers hold things smaller than themselves, that people act on their beliefs. Levesque, Davis and Morgenstern proposed the Winograd schema challenge as a test [7]. Each item is a pair of sentences that differ in one word, which flips the referent of a pronoun. One pair concerns a trophy that does not fit into a suitcase because it is too large or too small. The pronoun refers to the trophy in one case and to the suitcase in the other. Surface statistics that ignore the special word can do no better than chance.

Relevance in present systems

Large language models answer many Winograd-style questions correctly, which shows that statistics over vast text capture much commonsense knowledge. Whether they solve the relevance problem or sidestep it is debated. A model that has absorbed enormous numbers of situations may retrieve the relevant pattern without ever facing Dennett's explosion. It may also fail on situations that its training did not cover. The interactive lab reconstructs Dennett's robots with an adjustable time budget, and the Colab notebook counts frame axioms and tests surface heuristics on Winograd-style pairs.

> ***Pause and reflect.*** *When you decide what to check before leaving home, how do you avoid considering irrelevant facts? Could that procedure be written down?*

Hands-on components

Interactive lab: A robot must pull a wagon carrying its battery out of a room before a bomb explodes; the bomb is also on the wagon. Choose a strategy or pick the facts that the robot considers, then set the time before the explosion. Each inference costs one unit of time [2]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-08/lab.html

Colab notebook: It grounds a blocks world in STRIPS operators and counts the frame axioms that an explicit situation calculus encoding would need. It then tests two surface heuristics on Winograd-style sentence pairs [4, 5, 7]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-08/NB08_frame_problem.ipynb

Reflection prompts

1. In the interactive lab, how did you know which facts to tick? Could that knowledge be written as rules?
2. Do large language models solve the relevance problem, sidestep it or merely hide it? Give an example that supports your answer.
3. Describe a failure of common sense that you have seen in an AI system and explain it with the ideas of this week.

Weekly task and submission

What to do. Distinguish two versions of the frame problem with examples. First, explain the logical frame problem with the frame axioms of the Colab notebook. Second, explain the problem of relevance with Dennett's robots in the interactive lab. Third, assess whether present language models face the second problem, using one example from your field.

Where to look. Lecture: sections "A logical problem", "A problem of relevance" and "Relevance in present systems". Interactive lab: Dennett's robots and the bomb, where each

inference costs time (tab: Explore). Colab notebook: Section 2 with Exercise 1, which counts frame axioms. Section 3 with Exercise 2 tests surface heuristics on Winograd-style pairs.

Example. A planning assistant must update a project plan after a supplier is delayed. The logical problem asks how the system knows that the delay leaves the office address unchanged. The relevance problem asks how it finds the few tasks that the delay affects without checking every fact, which is the trap of R1D1 in Dennett's story.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] McCarthy, J., & Hayes, P. J. (1969). Some philosophical problems from the standpoint of artificial intelligence. In B. Meltzer & D. Michie (Eds.), Machine Intelligence 4 (pp. 463-502). Edinburgh University Press.

[2] Dennett, D. C. (1984). Cognitive wheels: The frame problem of AI. In C. Hookway (Eds.), Minds, Machines and Evolution (pp. 129-151). Cambridge University Press.

[3] Davis, E., & Marcus, G. (2015). Commonsense reasoning and commonsense knowledge in artificial intelligence. Communications of the ACM, 58(9), 92-103. https://doi.org/10.1145/2701413

[4] Fikes, R. E., & Nilsson, N. J. (1971). STRIPS: A new approach to the application of theorem proving to problem solving. Artificial Intelligence, 2(3-4), 189-208. https://doi.org/10.1016/0004-3702(71)90010-5

[5] Reiter, R. (2001). Knowledge in Action: Logical Foundations for Specifying and Implementing Dynamical Systems. MIT Press.

[6] Shanahan, M. (1997). Solving the Frame Problem: A Mathematical Investigation of the Common Sense Law of Inertia. MIT Press.

[7] Levesque, H. J., Davis, E., & Morgenstern, L. (2012). The Winograd schema challenge. In Proceedings of the 13th International Conference on Principles of Knowledge Representation and Reasoning (KR 2012) (pp. 552-561). AAAI Press.